

Assignment-06

1. Create a tuple with different data and create an empty tuple using built-in function.

```
In [1]: t1 = (1, True, 2.4, "Silicon")
        empt = tuple()

        print("The mixed tuple is: ", t1)
        print("Empty tuple is: ", empt)
```

```
The mixed tuple is: (1, True, 2.4, 'Silicon')
Empty tuple is: ()
```

2. Print the 3rd element from beginning and 4th element from end of a tuple.

```
In [7]: a = (10, 20, 30.5, True, 50, 60, 70)
        print("3rd element from beginning: ", a[2])
        print("4th element from end: ", a[-4])
```

```
3rd element from beginning: 30.5
4th element from end: True
```

3. Count the number of items present in a tuple and then add two new items into it (one item using concatenation operator and another item using built-in function).

```
In [8]: tup = (10, 20, 30)

        count = len(tup)
        print("Number of items in the tuple:", count)

        tup = tup + (40,)
        print("After concatenation:", tup)

        temp = list(tup)
        temp.append(50)
        tup = tuple(temp)
        print("After using built-in functions:", tup)
```

```
Number of items in the tuple: 3
After concatenation: (10, 20, 30, 40)
After using built-in functions: (10, 20, 30, 40, 50)
```

4. Find the number of occurrences for any repeated item present in a tuple.

```
In [11]: t = (10, 20, 30, 20, 40, 20, 50, 30)
n = int(input("Enter the item: "))

print("The occurrence is: ", t.count(n))
```

Enter the item: 20
The occurrence is: 3

5. Display the elements present in a tuple which are at the even position.

```
In [1]: t = (10, 20, 30, 40, 50, 60, 70, 80)

print("Elements at even position:")

for i in range(1, len(t), 2):
    print(t[i])
```

Elements at even position:
20
40
60
80

6. Remove the elements of a tuple which are present in the odd index position.

```
In [9]: t = (10, 20, 30, 40, 50, 60, 70, 80)

t = tuple(t[i] for i in range(len(t)) if i % 2 == 0)

print("Tuple after removing elements at odd index position:",t)
```

Tuple after removing elements at odd index position: (10, 30, 50, 70)

7. Remove the odd number elements from a tuple.

```
In [8]: t = (10, 15, 20, 25, 30, 35, 40, 45)

t = tuple(x for x in t if x % 2 == 0)

print("Tuple after removing odd numbers:",t)
```

Tuple after removing odd numbers: (10, 20, 30, 40)

8. Reverse the elements of a tuple by using and without using a built-in function.

```
In [7]: t = (10, 20, 30, 40, 50)

t = tuple(reversed(t))

print("Reversed tuple:",t)
```

Reversed tuple: (50, 40, 30, 20, 10)

9. Sort the elements present in a tuple in descending order.

```
In [6]: t = (10, 50, 20, 40, 30)

t = tuple(sorted(t, reverse=True))

print("Tuple in descending order:",t)
```

Tuple in descending order: (50, 40, 30, 20, 10)

10. Take a tuple of mixed data types and find the square of each element.

```
In [10]: t = (2, 4.5, 6, 8.5, 10)

for i in t:
    print(i, "square =", i ** 2)
```

2 square = 4
4.5 square = 20.25
6 square = 36
8.5 square = 72.25
10 square = 100

11. Given few numbers in a tuple, return 'True' if the first and last numbers are same.

```
In [11]: t = (10, 20, 30, 40, 10)

if t[0] == t[-1]:
    print(True)
else:
    print(False)
```

True

12. Print only those elements present in a tuple which are divisible by 5.

```
In [3]: t = (10, 12, 15, 23, 25, 31, 40, 47)
```

```
print("Elements divisible by 5:")  
for i in t:  
    if i % 5 == 0:  
        print(i)
```

Elements divisible by 5:

10

15

25

40

13. Given a tuple of numbers, create another tuple such that it contains only the even numbers of the first tuple.

```
In [2]: t = (10, 15, 22, 35, 40, 47, 50, 63)
```

```
new_tuple = ()
```

```
for i in t:  
    if i % 2 == 0:  
        new_tuple = new_tuple + (i,)
  
print("Tuple containing even numbers:", new_tuple)
```

Tuple containing even numbers: (10, 22, 40, 50)

14. Display the largest number of a tuple without using the built-in function max(). The user needs to input the values in tuple from the keyboard.

```
In [14]: a = int(input("Enter the no.of elements: "))
T1 = ()

for i in range(a):
    ele = int(input("Enter the elements:"))
    T1 = T1 + (ele,)

largest = T1[0]

for i in T1:
    if i > largest:
        largest = i

print(T1)
print("Largest element =", largest)
```

```
Enter the no.of elements: 5
Enter the elements: 25
Enter the elements: 10
Enter the elements: 45
Enter the elements: 30
Enter the elements: 15
```

```
(25, 10, 45, 30, 15)
Largest element = 45
```

15. Replace the last value of the tuple with 100. Sample list: [(1, 2, 3), (4, 5, 6), (7, 8, 9)] Output: [(1, 2, 100), (4, 5, 100), (7, 8, 100)]

```
In [15]: T1 = [(1, 2, 3), (4, 5, 6), (7, 8, 9)]
T2 = []

for i in T1:
    T2.append(i[:-1] + (100,))

print(T2)
```

```
[(1, 2, 100), (4, 5, 100), (7, 8, 100)]
```

16. Sort a tuple by its float element. Sample data: [('item1', '12.20'), ('item2', '15.10'), ('item3', '24.5')] Output: [('item3', '24.5'), ('item2', '15.10'), ('item1', '12.20')]

```
In [4]: T1 = [('item1', '12.20'), ('item2', '15.10'), ('item3', '24.5')]

T1 = sorted(T1, reverse=True)
print(T1)
```

```
[('item3', '24.5'), ('item2', '15.10'), ('item1', '12.20')]
```

17. Count the number of elements in a tuple until an element is a list.

```
In [17]: T1 = (10, 20, 30, [40, 50], 60, 70)
count = 0

for i in T1:
    if type(i) == list:
        break
    count = count + 1

print("Number of elements =", count)
```

Number of elements = 3

18. Swap the two tuples: tuple1 = (11, 22) tuple2 = (99, 88) Output: tuple1 = (99, 88) tuple2 = (11, 22)

```
In [18]: tuple1 = (11, 22)
tuple2 = (99, 88)

tuple1, tuple2 = tuple2, tuple1

print("tuple1 =", tuple1)
print("tuple2 =", tuple2)

tuple1 = (99, 88)
tuple2 = (11, 22)
```